

can be seen in [4, 6, 12]. Throughout the paper, ‘increasing’ means ‘non-decreasing’ and ‘decreasing’ means ‘non-increasing’. Whenever we use a derivative, an expectation or a conditional distribution, we are tacitly assuming that they exist.

If $X, Y \in \mathcal{L}_1$, then we say that:

- X is less than Y in the stochastic order (or in the first-order stochastic dominance), shortly written as $X \leq_{ST} Y$, if $\bar{F}_X \leq \bar{F}_Y$ or, equivalently, if $E(u(X)) \leq E(u(Y))$ for all increasing functions u such that these expectations exist. Then $X =_{ST} Y$ represents equality in law (distribution).
- X is less than Y in the increasing concave order (or in the second-order stochastic dominance), shortly written as $X \leq_{ICV} Y$, if $E(u(X)) \leq E(u(Y))$ for all increasing concave functions u such that these expectations exist or, equivalently, if

$$\int_{-\infty}^t F_X(x)dx \geq \int_{-\infty}^t F_Y(x)dx \quad \text{for all } t.$$

- X is less than Y in the increasing convex order (or in the stop-loss order), shortly written as $X \leq_{ICX} Y$, if $E(u(X)) \leq E(u(Y))$ for all increasing convex functions u such that these expectations exist or, equivalently, if

$$\int_t^{\infty} \bar{F}_X(x)dx \leq \int_t^{\infty} \bar{F}_Y(x)dx \quad \text{for all } t.$$

The function $\pi(t) = \int_t^{\infty} \bar{F}(x)dx$ is called *stop-loss function* (see e.g. [6], p. 19). Hence, $X \leq_{ICX} Y$ holds if and only if $\pi_X \leq \pi_Y$. It is also equivalent to $-X \geq_{ICV} -Y$.

- X is less than Y in the convex order, shortly written as $X \leq_{CX} Y$, if $E(u(X)) \leq E(u(Y))$ for all convex functions u such that these expectations exist or, equivalently, if $\pi_X \leq \pi_Y$ and $\lim_{t \rightarrow -\infty} \pi_X(t) - \pi_Y(t) = 0$. It implies $E(X) = E(Y)$ and $Var(X) \leq Var(Y)$. It is also equivalent to $X \leq_{ICX} Y$ and $E(X) = E(Y)$ (see [6], p. 19) and to $X \geq_{ICV} Y$ and $E(X) = E(Y)$.
- X is less than Y in the convolution order, shortly written as $X \leq_{CONV} Y$ if $Y =_{ST} X + Z$ where Z is a nonnegative random variable independent of X (see [12], p. 70 and [13]).

The relationships between these orders are summarized as follows:

$$\begin{array}{ccccccc} X \leq_{CONV} Y & \Rightarrow & X \leq_{ST} Y & \Rightarrow & X \leq_{ICX} Y & \Leftarrow & X \leq_{CX} Y \\ & & \Downarrow & & \Downarrow & & \Downarrow \\ X \geq_{CX} Y & \Rightarrow & X \leq_{ICV} Y & \Rightarrow & E(X) \leq E(Y) & & E(X) = E(Y) \end{array}$$